PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: IndoRoute AI

Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysian sellers face a "manual gap" when expanding to Indonesia. Unlike other regions, Indonesia requires self-managed translations, constant monitoring of volatile IDR currency, and strict adherence to BPOM (Food & Drug) safety laws. Mistakes lead to seized goods and destroyed profits.
- **Target Domain:** Domain 2 - AI for Economic Empowerment & Decision Intelligence.
- **Proposed Solution Summary:** IndoRoute AI is a strategic assistant using the Z.AI GLM Brain to automate cross-border safety checks. It simplifies the process by reading local regulations, protecting margins from exchange rate drops, and predicting customs delays. It helps business owners scale safely into Southeast Asia's largest market.

2. Background & Business Objective

- **Background of the Problem:** For years, selling to Indonesia has been a "black box" for Malaysian merchants. While basic inventory tools exist, they fail to alert sellers to sudden customs changes or regulatory updates. Consequently, sellers often ship blindly, leading to goods being stuck in 30-day inspections, resulting in mass refunds and financial loss.
- **Importance of Solving This Issue:** By automating the "safety checks" for cross-border trade, we can protect local sellers from losing thousands of ringgit in seized inventory. It lowers the technical and legal barriers, allowing more Malaysians to tap into Southeast Asia's largest consumer market with confidence.
- **Strategic Fit / Impact:** It aligns with the goal of using AI to empower small businesses. By integrating Z.AI for document parsing and decision-making, we turn a high-risk manual process into a safe, automated workflow.

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3. Product Purpose

3.1. Main Goal of the System: To act as a reliable partner that helps Malaysian sellers navigate the Indonesian market safely and profitably.

3.2. Intended Users (Target Audience):

- Small Business Owners and Entrepreneurs
- Logistics Managers and Cross-border Aggregators
- Trade Agencies & Business Consultants

4. System Functionalities

4.1. Description: The system operates as a decision engine that monitors Indonesian market data and seller inventory to provide actionable alerts.

4.2. Key Functionalities:

- **Compliance Radar:** Scans product titles and descriptions to ensure they meet Indonesian BPOM and legal standards**.**
- **Dynamic Profit Shield:** Monitors exchange rates and suggests price updates to maintain profit margins
- **Policy Sentinel:** Reads local news and official PDF documents to warn users of upcoming customs "Red Light" periods.
- **Logistics Router:** Recommends the most cost-effective shipping method (direct mail vs. local warehouse) based on current congestion levels.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: We utilize the Z.AI GLM for its high-context window (100k tokens), which is essential for parsing long Indonesian regulatory PDFs, and its native understanding of Bahasa Indonesia nuances.

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4.3.2. Prompting Strategy: The system uses "Few-Shot" examples to ground the AI. We provide the GLM with examples of illegal terms (e.g., *"Cure"*) vs. legal terms (e.g., *"Soothe"*) and enforce a Structured JSON Output. This ensures the data can be parsed by our FastAPI backend without conversational "filler" text.

4.3.3. Context & Input Handling: The system can take in about 100,000 tokens at once. For very long Indonesian legal documents that go over this limit, we break the PDF into smaller, overlapping parts. This prevents the system from 'losing its place' or cutting off important safety clauses at the end of a document.

4.3.4. Fallback & Failure Behavior:

- **Logic:** If the Z.AI GLM returns a response that does not match our JSON schema or has a "Confidence Score" below 0.8, the system triggers a Graceful Error State.
- **Action:** Instead of displaying potentially hallucinated legal advice, the UI will show a "Human Review Required" flag. This prevents the seller from making a high-risk shipping decision based on uncertain AI output.

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5. User Stories & Use Cases

- **User Stories:**

  - **Story 1:** "As a Malaysian seller, I want the AI to summarize Bahasa Indonesia news alerts so I can identify unannounced 'Red Light' periods before I ship."

  - **Story 2:** "As an entrepreneur, I want my product descriptions checked for BPOM 'red-flag' words so my goods are not seized by Jakarta customs."

  - **Story 3:** "As a small business owner, I want to see how currency fluctuations affect my Ringgit profit so I can adjust prices instantly."

  - **Story 4:** "As a logistics manager, I want the system to recommend a shipping route based on current port congestion so my delivery times remain consistent."

  - **Story 5:** "As a business owner looking to scale, I want the system to provide a prioritized 'Action Command' that weighs compliance risks against profit margins, so I can decide whether to pause shipping a specific SKU or switch to a local warehouse without manually calculating the trade-offs."

- **Use Cases (Main Interactions):**

  Policy & Compliance Scan Flow: Users upload a product listing or a regulatory PDF. IndoRoute AI parses the text to identify specific risk factors, such as prohibited "Medical Grade" claims, and then renders a "Compliance Score" with suggested edits to avoid border seizure.

  Dynamic Profit Protection Flow: The system monitors live IDR/RM exchange rates. When a significant currency drop is detected, the AI generates an updated pricing strategy for the user's Indonesian store while maintaining the original profit margin in Malaysian Ringgit.

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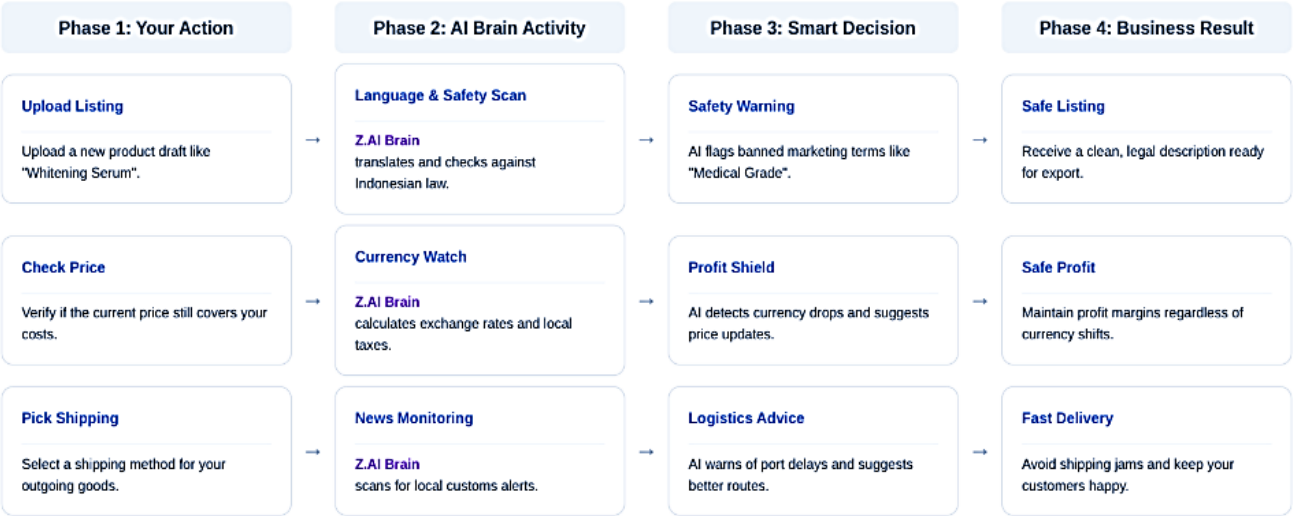
Intelligent Logistics Routing: Users enter a target SKU and delivery timeline. IndoRoute AI triggers a "Trade-off Computation" that compares current Direct Mail risks against Overseas Warehouse costs, recommending the safest route to avoid unannounced customs delays.

- **Use Case Diagram:**



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6. Features Included (Scope Definition)

- **AI-Powered Compliance Audit:** Automated scanning of product listings against a grounded database of BPOM (Indonesian FDA) regulations using the Z.AI GLM.
- **Dynamic Profit Shield:** Real-time calculation of net profit margins that accounts for MYR/IDR exchange rates, Indonesian import duties (BM, PPN, PPH), and logistics fees.
- **Predictive Trade Routing:** A decision engine that analyzes customs news alerts (Sentinel) to recommend shipping adjustments (e.g., switching from Direct Mail to Jakarta Warehousing during "Red Light" periods).

7. Features Not Included (Scope Control)

- **Direct Payment Gateway Integration:** The system will provide financial advice and price suggestions but will not handle actual currency exchange or bank transfers.
- **Official Legal Representation:** The AI provides risk scores and compliance suggestions; it is not a substitute for official legal counsel or government-authorized certification services.
- **Real-Time Inventory Synchronisation:** While the system tracks 50+ SKUs for the demo, it will not integrate with live warehouse management systems (WMS) or Shopee Seller Centre API for live inventory counts.

8. Assumptions & Constraints

- **Token Management:** To keep costs low, the system will cache common regulatory summaries so the AI doesn't have to re-read the same law for every user.
- **Cost & Speed Optimization:** To keep the system fast and affordable, we use a "results library." If multiple entrepreneurs are selling similar items (like skincare or electronics), the system retrieves previously verified safety standards for that category instead of starting a new AI analysis. This keeps our token costs manageable and ensures users get instant answers.

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- **Technical Constraints:** The system relies on the availability of public data feeds from Indonesian customs and the Shopee/Lazada API.
- **User Input:** The system provides recommendations; final approval for price changes or shipping remains with the human seller.

9. Risks & Questions Throughout Development

- **Regulation Speed:** How quickly can the AI process a new law once it is published?
- **Data Accuracy:** What happens if the news source for customs alerts is unreliable?
- **Language Nuance:** Could the AI misunderstand local Indonesian slang or specific trade terms?

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